

Model Collapse on Manifolds: a Closed-Form Capacity Floor and a Data-Anchoring Fix

Naman Chandwani
Independent Researcher
namanchandwani.10@gmail.com

July 2026

Abstract

Generative models are increasingly trained on data that includes their own past outputs, and for diffusion models this *self-consuming* loop is known to degrade quality from one generation to the next. The sharpest existing theory [1] shows that a carefully annealed early-stopping schedule repairs the loop—but only when the data density is smooth, explicitly leaving open the case that matters most in practice, where data concentrate on a thin low-dimensional manifold. We settle this case, in two parts. *Negatively*, we prove a *capacity floor*: near a manifold of width σ the score a diffusion model must represent is steep only inside a short, finite window of noise levels, so any network whose slope is capped below that peak bakes in a fixed amount of excess width on every pass that later denoising cannot remove. In the singular limit the floor is *closed form*— $1/(8\bar{\kappa}^2)$ in the network’s slope cap $\bar{\kappa}$, with no dependence on the sampling schedule—so a fixed model collapses without bound as the manifold thins, and staying bounded requires capacity growing like $\sqrt{J}$ in the Fisher information J . Empirically the trap is two-horned: the stochastic sampler sticks $\sim 11\times$ too wide, while the deterministic one collapses to a point; neither reaches the data. *Constructively*, we prove that periodically re-matching the model’s output to a small, *fixed* pool of real examples—matching their *local width*, not re-inserting the examples themselves—makes the loop *memoryless*, so error stops compounding for every positive fraction of real data. Over a 20-generation head-to-head the prior state of the art plateaus at $9.8\sigma^2$ while anchoring holds $1.01\sigma^2$, and on pixel-space MNIST anchoring removes $\sim 81\%$ of the off-manifold collapse. All experiments run on CPU and are reproducible from released scripts.

1 Introduction

Web-scale corpora already contain machine-generated data, and synthetic data is used deliberately in modern training pipelines. When a generative model is repeatedly retrained on a pool containing its own previous outputs, quality degrades across generations—*model collapse* [2–5]. For diffusion models, Khelifa et al. [1] give the sharpest analysis to date: with a fraction λ of fresh real data per generation and an annealed truncation schedule $t_0(g) \rightarrow 0$, the recursion converges back to the data distribution—*provided the data has finite Fisher information* (their Case 1). They explicitly leave open their Case 2: data supported on, or concentrated near, a low-dimensional manifold, where the Fisher information $J \asymp (d - k)/\sigma^2$ diverges as the noise width $\sigma \rightarrow 0$. Essentially all natural data—images, audio, molecular conformations—is of this second kind [6, 7].

This paper answers Case 2.

The idea in one paragraph. When a model repeatedly learns from its own samples, the hardest property to preserve is the data’s *thinness*—the exact width of a pen stroke, the sharp edge of a shape. We show that a finite-capacity denoiser can reproduce that thinness only inside a brief window of noise levels, so a small, fixed amount of blur is baked in every generation and accumulates. Two things follow. The blur has an *exact floor* set by a single number—the steepest slope the network can represent—and *no* training or sampling schedule can lower it. And the accumulation can be cancelled by periodically re-matching the model’s output to a small, unchanging set of real examples: not by copying the examples back into training, but by rescaling to *their local width*. The rest of this section makes both statements precise—the first is our *negative* result (a capacity floor), the second our *constructive* one (a data-anchoring fix).

The negative half is a *capacity floor* (Section 5). The score that a diffusion model must represent near a σ -thin manifold has normal slope $\kappa(t) = s(t)/\gamma^2(t)$ which is *non-monotone* in diffusion time: it peaks at $\kappa_{\max} = 1/(2\sigma)$ near $t \approx \sigma^2$ and falls back toward zero as $t \rightarrow 0$ (Figure 1a). A network whose normal slope is bounded by $\bar{\kappa}$ therefore under-contracts on a *finite band* of diffusion times; the excess variance frozen in while crossing that band can never be removed afterwards, because too few contraction e-folds remain below the band. We prove that in the singular limit the resulting floor is a universal function of the single ratio $\rho = \bar{\kappa}/\kappa_{\max}$ and admits *closed forms*: under no assumption beyond the slope bound, the floor is exactly $\sigma^2/(2\rho^2) = 1/(8\bar{\kappa}^2)$ —so a fixed network collapses relatively without bound as $\sigma \rightarrow 0$, and bounded collapse requires $\bar{\kappa} = \Omega(1/\sigma) = \Omega(\sqrt{J})$. No truncation schedule enters these statements: annealing targets a term the schedule can kill, while the floor lives in a term it cannot touch. Empirically the trap is *two-horned* (Figure 2): keeping the reverse-SDE noise, the sampler sticks $\sim 11\times$ too wide; removing it, the recursion collapses to a single point. Neither branch reaches the data.

The constructive half is an anchoring operator (Section 6). A bidirectional, per-axis, local moment match of the training pool (and sampler output) to a small *stale* reference of real data—2,000 samples drawn once at generation zero—pins the normal variance at the reference value each generation. We prove this makes the recursion *memoryless*: the fixed-point equation loses its dependence on the previous generation, the critical fraction disappears, and the dynamics of Khelifa et al. [1]’s Theorem 4.1 are restored in the singular regime. In a 20-generation head-to-head against their annealed-truncation remedy (Figure 5), the annealed arm plateaus at $9.8\sigma^2$ while the anchored arms hold $1.01\sigma^2$, flat, on both samplers we test; on real pixels (MNIST), anchoring closes $\sim 81\%$ of the collapse gap (Figure 6).

Contributions.

1. **A law of collapse** (Theorem 1): under a measured one-dimensional concave response law, the variance recursion has a unique plateau for $\lambda > \lambda^* = c_\infty/(1 + c_\infty)$ and no plateau below; truncation enters only additively, which makes precise why annealing schedules cannot help singular data.
2. **A closed-form capacity floor** (Theorem 2, Corollary 1): any reverse-SDE sampler with normal score slope $\leq \bar{\kappa}$ is floored; in the singular limit the floor is $1/(8\bar{\kappa}^2)$ unconditionally and $((1 + 3e^{-4})/8)\bar{\kappa}^{-2}$ for the empirically observed no-overshoot class—two constants that differ by only 5.5%. Bounded relative collapse as $\sigma \rightarrow 0$ requires $\bar{\kappa} \geq 1/(2\sigma\sqrt{2C})$.
3. **A memorylessness theorem for anchoring** (Theorem 3): matching pool and output normal variance to a stale real reference removes the recursion’s memory, for every $\lambda > 0$, with an error bound independent of the ambient dimension.

4. **Falsifiable empirics:** a zero-fit prediction of the measured sampling floor from one measured network property; a two-horned ablation separating the floor’s existence (sampler noise) from its size (slope ceiling); a 20-generation, 5-seed head-to-head win over the state-of-the-art remedy; and a pixel-space MNIST recursion.

Throughout we keep the epistemic status of each claim explicit: *we prove* (mathematics, given stated assumptions), *we observe* (multi-seed experiments), or *we model* (measured structural laws that we do not derive). Section 8 collects everything in the third category.

2 Related work

Model collapse. Collapse under recursive training is documented and analyzed across model families [2–5, 8]; accumulation (rather than replacement) of data and sufficient fresh-data fractions mitigate it in the absolutely continuous setting. Khelifa et al. [9] bound error propagation in recursively trained diffusions via a discounted accumulation argument, but their assumptions require absolutely continuous data—structurally excluding manifolds—with fixed truncation and upper bounds rather than laws. Khelifa et al. [1] characterize the limiting collapse distribution spectrally and prove annealed truncation converges for finite-Fisher data (their Case 1), stating manifold-supported data (Case 2) as open. We give the Case 2 answer: a floor theorem for their sampler class, and an operator restoring their Case 1 dynamics.

Diffusion on manifold data. That diffusion models detect and concentrate on data manifolds is established [7, 10]. The singular ($\sigma \rightarrow 0$) score is the object of a growing one-shot literature: optimal denoising under low regularity [11], provable learning under the manifold hypothesis [12], and mitigation of the score singularity [13]. These works are one-generation; our subject is the *recursion*, where a per-generation floor compounds into a permanently displaced plateau. We use the Tweedie/posterior-mean jump [11, 14] as a raw-error reducer inside the loop and are explicit that it does not, by itself, break the floor (Section 6).

Local geometry estimation. Our anchoring operator estimates local tangent frames by k NN-PCA; its error analysis imports standard local-PCA perturbation rates [15, 16]. The operator needs only *shared* frames between pool and reference, not correct ones, which is why these rates enter only at second order (Lemma 5).

3 Setup

Data model (σ -tube). Let $M \subset \mathbb{R}^d$ be a compact smooth k -dimensional manifold with reach $\tau > 0$. Data are $X_0 = \varphi(U) + \sigma N_\perp$, with U uniform on M and $N_\perp$ standard normal in the $(d - k)$ -dimensional normal space; $\sigma > 0$ is the tube width. The singular regime is $\sigma \rightarrow 0$; the Fisher information of the marginal scales as $J \asymp (d - k)/\sigma^2 \rightarrow \infty$ (Case 2 of 1).

Forward process and the normal coordinate. We use the variance-preserving OU forward process $X_t = a(t)X_0 + s(t)Z$ with $a(t) = e^{-t/2}$, $s^2(t) = 1 - e^{-t}$ [17, 18]. Fix a point of M and work in a normal coordinate $x \in \mathbb{R}$, treating the fiber marginal as Gaussian and decoupled from tangential position (*flat-tube idealization*; curvature enters at $O(s^2/\tau)$ and is tracked where it matters). Then $X_0 \sim \mathcal{N}(0, \sigma^2)$, $X_t \sim \mathcal{N}(0, \gamma^2(t))$ with $\gamma^2 = a^2\sigma^2 + s^2$, and the denoising target is linear:

$$\varepsilon^*(x, t) = \kappa(t)x, \quad \kappa(t) = \frac{s(t)}{\gamma^2(t)}, \quad \nabla \log p_t(x) = -\frac{x}{\gamma^2(t)}. \quad (1)$$

κ is the normal slope the score network must represent. It satisfies $\kappa \approx 1/s$ for $s^2 \gg \sigma^2$, peaks at $\kappa_{\max} = 1/(2\sigma)$ near $t \approx \sigma^2$, and falls back to 0 as $t \rightarrow 0$ (Figure 1a). In the singular limit the *peak* slope diverges—the precise sense in which infinite Fisher information bites the estimator.

Samplers. $S_{\text{trunc}}(t_0)$ integrates the learned reverse SDE from T down to $t_0 > 0$ (Euler–Maruyama, $x \leftarrow x + (-\frac{1}{2}x + \hat{\varepsilon}/s)dt + \sqrt{|dt|}\xi$) and outputs x_{t_0} ; all prior remedies, including the annealed schedules of Khelifa et al. [1], are of this class. $S_{\text{jump}}(\hat{t})$ integrates only to a moderate $\hat{t}$ and then outputs the Tweedie posterior-mean jump $\hat{x}_0 = (x_{\hat{t}} - s(\hat{t})\hat{\varepsilon})/a(\hat{t})$ [11, 14].

The recursion. For a distribution p on $\mathbb{R}^d$ define the per-direction normal second moment $v(p) = \mathbb{E}_p[\text{dist}(X, M)^2]/(d - k)$; true data has $v = \sigma^2$. Generation g trains $\hat{\varepsilon}_g$ by denoising score matching on $\text{pool}_g = \lambda(\text{fresh real}) + (1 - \lambda)(\text{generation } g-1)$ and samples; v_g is the output value. Since v is linear in the measure, $w_g := v(\text{pool}_g) = \lambda\sigma^2 + (1 - \lambda)v_{g-1}$ exactly.

Modeling assumptions (*we model*). Our law rests on two assumptions about the trained pipeline that we *measure and cross-validate* but do not derive.

Assumption 1 (scalar reduction). *The per-generation map factors through the pool width: $v_g = F_g(w_g)$. Why a scalar: the collapse observable is a variance along the normal fibre—one number per normal axis by construction—so this is an order-parameter statement (the width closes on itself), not a claim that the network is scalar, in the same way an order parameter closes a mean-field recursion without the microscopic state being low-dimensional. Validation: fixed points measured in one experiment type predict those of an independent type within 1–13% across six (t_0, λ) cells (Section 7).*

Assumption 2 (concave response). $F_g(w) = a^2(t_0(g))w + s^2(t_0(g)) + e^2(w)$, where the sampling excess $e^2 : [0, \infty) \rightarrow (0, \infty)$ is nondecreasing and concave with $e^2(0) = e_0^2 > 0$. Validation: the measured local slope of e^2 runs from +0.6 (fat pools) to ≈ 0 (thin pools)—concave and saturating; its small magnitude is itself a measured near-cancellation of two $O(0.7)$ channels (Section 7).

4 The law of collapse

Under Assumptions 1–2 the recursion is one-dimensional:

$$v_g = T(v_{g-1}) + s_g^2, \quad T(v) = \lambda\sigma^2 + (1 - \lambda)v + e^2(\lambda\sigma^2 + (1 - \lambda)v), \quad (2)$$

with $s_g^2 := s^2(t_0(g))$ the truncation remainder. Let $c_\infty := \lim_{w \rightarrow \infty} e^{2'}(w)$ (exists by concavity) and define the *critical real-data fraction* $\lambda^* := c_\infty/(1 + c_\infty)$.

Theorem 1 (law of the collapse; *we prove*). *Fix $\lambda \in (0, 1]$. (i) If $\lambda > \lambda^*$, T has a unique fixed point $v_\infty > 0$ solving $v = \lambda\sigma^2 + (1 - \lambda)v + e^2(\lambda\sigma^2 + (1 - \lambda)v)$, and for every $v_0 \geq 0$ and every truncation schedule with $s_g^2 \rightarrow 0$ the iterates converge, $v_g \rightarrow v_\infty$, with local rate $\rho_{\text{loc}} = (1 - \lambda)(1 + e^{2'}(w_\infty))$. (ii) If $(1 - \lambda)(1 + c_\infty) > 1$, then $T(v) - v \geq T(0) > 0$ for all v and $v_g \rightarrow \infty$: no plateau, for any schedule. (iii) In the affine special case $e^2(w) = e_0^2 + cw$: $\lambda^* = c/(1 + c)$, $\rho_{\text{loc}} = (1 + c)(1 - \lambda)$, and $v_\infty = ((1 + c)\lambda\sigma^2 + e_0^2)/(\lambda(1 + c) - c)$.*

Proof. Appendix A. Elementary but complete: concavity gives a unique downcrossing of $T(v) - v$; contraction above the fixed point plus a monotone barrier below give convergence under vanishing perturbations. $\square$

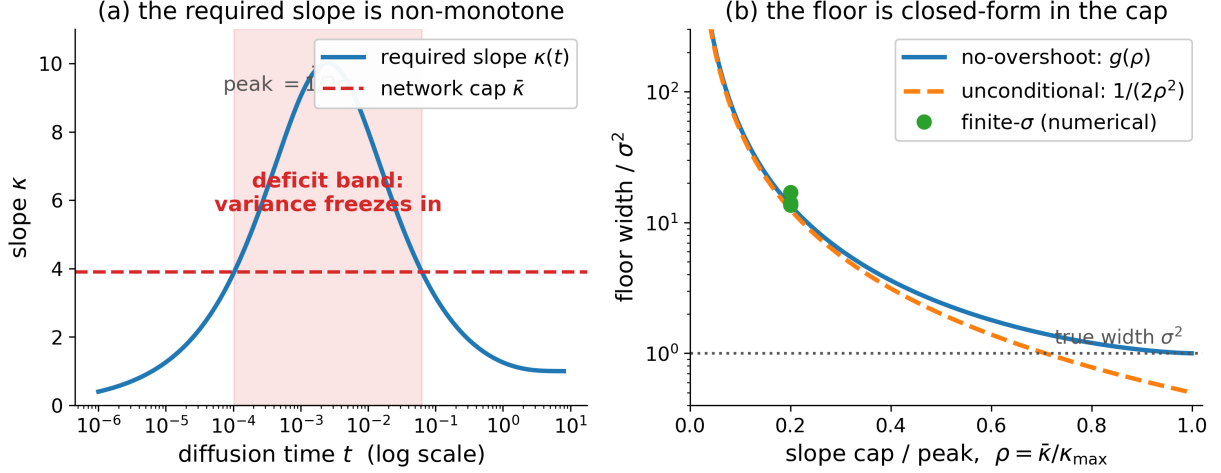


Figure 1: **The mechanism and the law.** (a) The required normal slope $\kappa(t) = s/\gamma^2$ is non-monotone: it peaks at $\kappa_{\max} = 1/(2\sigma)$ near $t \approx \sigma^2$ and falls back to zero. A network with ceiling $\bar{\kappa} < \kappa_{\max}$ under-contracts only on a *finite band* (shaded); variance frozen in the band cannot be removed below it (too few contraction e-folds remain). (b) The resulting floor in units of σ^2 as a function of $\rho = 2\sigma\bar{\kappa}$: closed-form no-overshoot law $g(\rho)$ (blue, Theorem 2), exact unconditional law $1/(2\rho^2)$ (orange), finite- σ ODE values converging to the limit (green markers at $\rho = 0.2$).

Remark 1 (why annealing cannot help singular data). Truncation enters (2) only through the additive, schedule-killable s_g^2 . The plateau v_∞ is determined by e^2 alone, which no schedule touches. The remedy of Khelifa et al. [1] is the special case $e^2 \equiv 0$ (their Case 1 assumptions make the estimation term schedule-dominated), where indeed $v_\infty = \sigma^2$. Case 2 is precisely the regime where e^2 has a floor—the subject of the next section.

5 The capacity floor

We now prove that for manifold data the sampling excess e^2 of any truncated reverse-SDE sampler is floored by network capacity. Write the trained ε -predictor’s affine part as $\hat{\varepsilon}(x, t) = \hat{\kappa}(t)x + b(t) + \delta(x, t)$, where $(\hat{\kappa}, b)$ is the affine $L^2(p_t)$ fit of the estimator itself and δ the orthogonal residual. Two structural facts (proofs in Appendix B):

Lemma 1 (variance ODE; we prove). *Under the reverse Euler scheme with affine estimator, the ensemble normal variance obeys, in the small-step limit,*

$$-\frac{dV}{dt} = \left(1 - \frac{2\hat{\kappa}(t)}{s(t)}\right)V + 1. \quad (3)$$

With the exact slope $\hat{\kappa} = \kappa$, $V(t) = \gamma^2(t)$ solves (3) exactly. The forcing +1 is the sampler’s own injected noise.

Lemma 2 (orthogonality shield; we prove, unconditional). *For any estimator, the residual δ satisfies $\mathbb{E}_{p_t}[\delta] = \mathbb{E}_{p_t}[x\delta] = 0$ (normal equations), hence has no first-order effect on (3): at first order the residual can only add variance. Thinning can enter only through the affine part.*

By Lemma 2, lower bounds derived from (3) with the affine part alone bound the full estimator. This is the precise sense in which the floor is *not* an artifact of linearizing the score: a real score network is highly nonlinear, but its nonlinearity is carried entirely by the orthogonal residual δ , which by the normal equations has zero first-order effect on the variance and can only *raise* it, so the affine lower bound is a lower bound for the fully nonlinear network. The crux is the geometry of the required slope:

Lemma 3 (finite band; we prove). $\kappa(t)$ rises from 0, peaks at $\kappa_{\max} = 1/(2\sigma)$ at $t \approx \sigma^2$, and falls back to 0 as $t \rightarrow 0$. Hence for any ceiling $\bar{\kappa} < \kappa_{\max}$ the deficit set $\{t : \kappa(t) > \bar{\kappa}\}$ is a finite band; below it the exact slope is representable again, but only $2\log(1 + \rho^2/4) \leq \rho^2/2$ contraction e-folds remain ($\rho := 2\sigma\bar{\kappa}$), so variance frozen in the band persists to $t_0 \rightarrow 0$.

In the singular limit the whole flow collapses onto a one-parameter family (Appendix C for proofs and error terms):

Lemma 4 (singular-limit ODE; we prove). Rescale $t = \sigma^2 u$, $V = \sigma^2 W$, and write the slope envelope as $\hat{\kappa}(t) = (1/\sigma)m(u)$ with $m(u) = \rho/2$ (unconditional envelope) or $m(u) = \min\{\sqrt{u}/(1+u), \rho/2\}$ (no-overshoot envelope). Then (3) becomes, as $\sigma \rightarrow 0$ at fixed ρ ,

$$-\frac{dW}{du} = 1 - \frac{2m(u)}{\sqrt{u}} W, \quad (4)$$

a σ -free equation; the uncapped equation has exact solution $W = 1 + u$ (the rescaled true marginal), and deviations contract quadratically under downward integration. Consequently $\lim_{t_0 \rightarrow 0} V(t_0)/\sigma^2 = g_{\text{class}}(\rho) + O(\sigma^2/\rho^2)$: **universality in ρ is a theorem**.

Theorem 2 (closed-form capacity floor; we prove). Let the trained score's normal slope satisfy $\hat{\kappa}(t) \leq \bar{\kappa} < \infty$ (finite normal Lipschitz constant—automatic for any finite network), and set $\rho = 2\sigma\bar{\kappa}$. Then for every truncation time $t_0 > 0$ and every schedule:

(U) **Unconditional**. With no further assumption (arbitrary overshoot permitted),

$$V(t_0) \geq \sigma^2 \frac{1}{2\rho^2} (1 + o(1)) = \frac{1}{8\bar{\kappa}^2} (1 + o(1)) \quad (\sigma \rightarrow 0), \quad (5)$$

where $1/(2\rho^2)$ is the exact value of (4) with $m \equiv \rho/2$: $W(0^+) = \int_0^\infty e^{-2\rho\sqrt{u}} du = 1/(2\rho^2)$. The floor exceeds σ^2 whenever $\rho < 1/\sqrt{2}$, i.e. $\bar{\kappa} < \kappa_{\max}/\sqrt{2}$.

(N) **No-overshoot**. If additionally $\hat{\kappa}(t) \leq \kappa(t)$ (the empirically observed profile; the network never over-contracts), the tighter floor $V(t_0) \geq \sigma^2 g(\rho)(1 + o(1))$ holds with the explicit closed form: writing $q = \sqrt{1 - \rho^2}$, $x_\mp = (1 \mp q)/\rho$, $u_\mp = x_\mp^2$,

$$W_{\text{exit}} = (1 + u_+)e^{-4q} + \frac{(3 - 2q) - (3 + 2q)e^{-4q}}{2\rho^2}, \quad g(\rho) = 1 + \frac{W_{\text{exit}} - (1 + u_-)}{(1 + u_-)^2}. \quad (6)$$

$g(\rho) > 1$ for every $\rho \in (0, 1)$ with $g \rightarrow 1$ only as $\rho \rightarrow 1^-$, and $g(\rho) = C_0/\rho^2 + O(1)$ as $\rho \rightarrow 0$ with $C_0 = (1 + 3e^{-4})/2 = 0.5275$, i.e. $\Phi_{\text{det}} \rightarrow ((1 + 3e^{-4})/8)\bar{\kappa}^{-2} = 0.1319\bar{\kappa}^{-2}$.

The two asymptotic constants, $1/8$ and $(1 + 3e^{-4})/8$, differ by 5.5%: the class distinction is quantitatively minor. The closed forms match direct integration of (4) to $\leq 2.4 \times 10^{-5}$ across $\rho \in [0.1, 1]$, and the finite- σ ODE converges to them from above at the stated $O(\sigma^2/\rho^2)$ rate.

Corollary 1 ($\sigma \rightarrow 0$: the answer to Case 2; we prove). Hold $\bar{\kappa}$ fixed and let $\sigma \rightarrow 0$ ($J \rightarrow \infty$). Then unconditionally $V/\sigma^2 \geq (1 + o(1))/(8\bar{\kappa}^2\sigma^2) \rightarrow \infty$: a fixed-capacity truncated sampler collapses

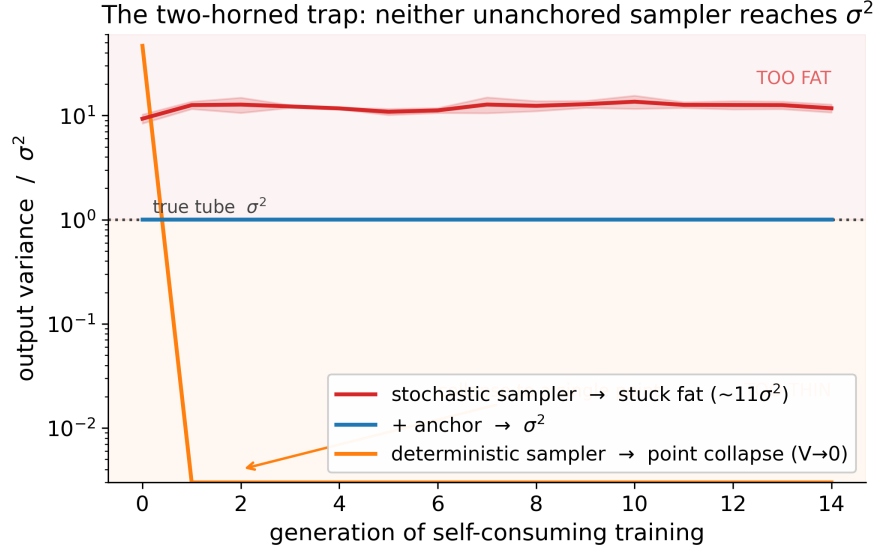


Figure 2: **The two-horned trap (*we observe*)**. Self-consuming recursions at fixed $t_0 = 10^{-4}$, no anchor, 3 seeds. With the reverse-SDE noise on, the recursion sticks fat at $10.8\text{--}13.2\sigma^2$ (red). With the noise zeroed (deterministic sampler), it collapses to a *single point* within one generation (orange; verified loss of all 2-D spread, no numerical overflow). Neither branch reaches the tube (σ^2 , dotted); the anchored recursion (blue, Section 6) holds it.

relatively *without bound*, for every annealing schedule. Conversely, bounded relative collapse $V/\sigma^2 \leq C$ requires

$$\bar{\kappa} \geq \frac{1}{2\sigma\sqrt{2C}} = \Omega(1/\sigma) = \Omega(\sqrt{J}). \quad (7)$$

Truncated sampling of singular data demands capacity growing with the singularity; no fixed architecture provides it.

What each ingredient does (*we observe*). The ablation of Figure 2 separates the roles exactly as (3) dictates: the sampler’s own noise (the +1) is load-bearing for the floor’s *existence*—switching it off sends $V \rightarrow 0$ (point collapse, the deterministic horn)—while the finite ceiling $\bar{\kappa}$ makes the floor *larger than* σ^2 (the stochastic horn). The two horns bracket the target from both sides; no setting of the noise knob reaches it.

Zero-fit confrontation with a real network (*we observe*). For our standard pipeline the measured slope profile tracks $\kappa(t)$ faithfully at benign times ($\hat{\kappa}/\kappa = 0.95\text{--}0.97$ for $t \geq 0.1$) and then saturates at $\bar{\kappa} \approx 3.9$ while κ continues to ~ 10 —an optimization ceiling, independent of sample size. Feeding $\bar{\kappa} = 3.9$ ($\rho = 0.39$) into the floor law predicts $\Phi_{\text{det}} = 3.8\sigma^2$ at $\sigma = 0.05$, matching the same network’s measured $t_0 \rightarrow 0$ sampling floor with *no free parameter* (Figure 3); integrating (3) with the full measured profile $\hat{\kappa}(t)$ reproduces 69–75% of the measured floor at each of three truncation times, with the exact-score control reproducing $\gamma^2(t_0)$ to all displayed digits. The remaining $\sim 30\%$ is the residual-field channel of Lemma 2, which we bound in a block-correlation model (*we model*; Appendix E).

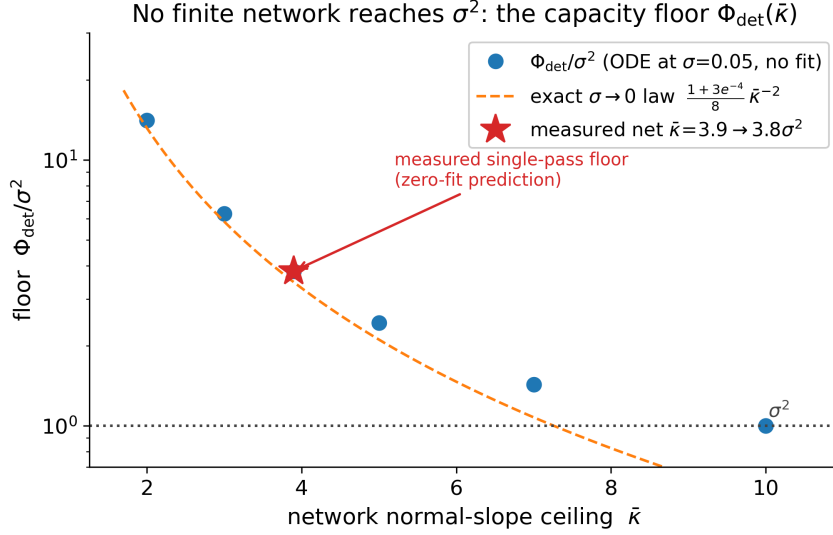


Figure 3: **The capacity floor, predicted and measured.** Blue: finite- σ variance-ODE floor at $\sigma = 0.05$ (no fit). Dashed: the exact $\sigma \rightarrow 0$ law $((1+3e^{-4})/8)\bar{\kappa}^{-2}$ of Theorem 2. Star: the measured network ceiling $\bar{\kappa} = 3.9$ predicts the measured sampling floor $3.8\sigma^2$ with zero free parameters.

An honest gap (*we model*). Iterating the floor through the recursion (every generation’s pool is at least as wide as $\lambda\sigma^2 + (1-\lambda)\Phi_{\text{det}}$) gives a deterministic fixed point $v_{\text{det}}^* = 4.5\sigma^2$ at our parameters. The measured recursion floor is $\approx 11\sigma^2$ —a factor 2.4 above. Our theorem claims only a *lower bound* on the recursion (respected), and we attribute the excess to per-generation compounding: a network trained on self-generated, fattened data has a worse slope profile than one trained on a clean tube of the same width, a channel the single-pass analysis deliberately omits. We leave the exact recursion constant open.

6 The anchoring fix

Corollary 1 says the singular regime cannot be won by scaling the model or tuning the schedule. It can be won by *anchoring the recursion to data*.

The operator. For each point, find its k nearest neighbors in the reference; take the local PCA frame (top- k_{dim} directions tangent, remainder normal); rescale the point’s normal coordinates so that the pool’s local normal second moments equal the *reference’s* local normal second moments in the *same estimated frames*. The match is *bidirectional*—it contracts fat directions and inflates thin ones—and *per-axis*. Both properties are forced by the theory: the sampler’s error is signed (fat from injection, thin from posterior-mean sharpening; Proposition 1 in Appendix D) and anisotropic whenever the normal spectrum straddles $s^2(\hat{t})$, so any scalar or one-sided matcher mis-corrects some directions (*we observe*: a scalar matcher leaves MNIST-latent thin ratios at 0.62; the per-axis matcher achieves $|\text{thin} - 1| \leq 0.012$).

Lemma 5 (shared-frame cancellation; *we prove* to first order). *Let both reference and pool normal energies be measured in the same k NN-PCA frames estimated from the reference. Then the matched pool satisfies, per normal direction, $v(\text{matched}) = \hat{\sigma}_{\text{ref}}^2 + O_p(m^{-1/2}) + \Delta_{\text{leak}}$, where the frame-error*

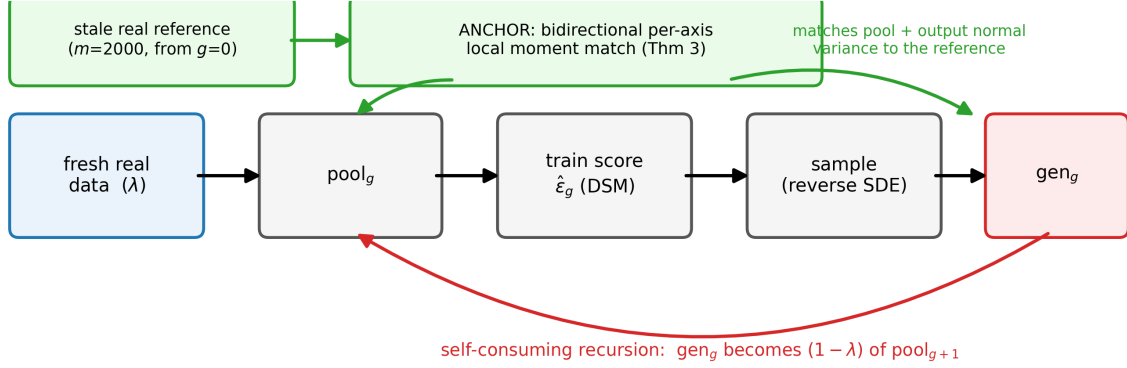


Figure 4: **Where the anchor sits.** Each generation, the pool (and, for the jump pipeline, the sampler output) is passed through a bidirectional per-axis local moment match against a *stale* reference of $m = 2,000$ real samples drawn once at $g = 0$. No provenance labels are needed; the reference is never refreshed.

leakage Δ_{leak} afflicts pool and reference identically at first order in the frame perturbation and cancels; standard local-PCA rates [15, 16] enter only at second order. Every quantity in the bound is intrinsic (k, τ, σ, m); the ambient dimension d appears nowhere.

Theorem 3 (anchoring makes the recursion memoryless; we prove given Lemma 5 and Assumption 1). *Insert the matcher after pooling (and after sampling, for the jump pipeline). Then the training input has normal variance $\hat{\sigma}_{\text{ref}}^2 + O(\text{err})$ regardless of v_{g-1} : the map (2) loses its v -argument,*

$$v_g = F_g(\hat{\sigma}_{\text{ref}}^2 + O(\text{err})),$$

*a constant sequence up to non-compounding $O(\text{err})$. Consequently: no fixed-point equation, **no critical fraction**—**stability for every $\lambda > 0$** ; and with S_{trunc} the dynamics reduce to truncation-only, restoring the hypotheses and conclusion of Khelifa et al. [1]’s Theorem 4.1 in the singular regime.*

Why this is not a replay buffer. Replay, data-accumulation, and data-refresh remedies [4, 8] fight collapse by *re-inserting real examples* into the pool: they preserve *samples*. The anchor preserves something weaker and more portable—the *local normal moments* (per-axis widths) of a small stale reference—and imposes them on *whatever the model currently generates*. It never adds a reference point to the training pool, uses no provenance labels, and reuses the same $m=2,000$ points for every generation. The distinction is not cosmetic. A replay buffer of size m still lets the $(1 - \lambda)$ synthetic majority set the pool’s fine-scale statistics, so the floor of Section 5 still governs that majority; anchoring instead *overwrites* the fine-scale statistics of the majority itself, which is why it removes the floor rather than diluting it. In the language of the law (Theorem 1): replay changes λ ; anchoring changes the response map F_g .

Attribution, stated plainly. In our pipeline the *matcher* is the floor-breaker; the Tweedie jump is a *raw-error reducer*. The jump’s raw output is itself fat ($7\text{--}20 \times \sigma^2$ across settings); its value is a $\approx 2\times$ smaller one-shot raw error and $\approx 3\times$ smaller unanchored plateau (0.052 vs 0.17 at $\lambda = \frac{1}{2}$, itself predicted by Theorem 1 as $e_{\text{jump}}^2/\lambda$), plus smaller post-match tails. Post-anchor, the standard and jump samplers converge to the same variance; we report both.

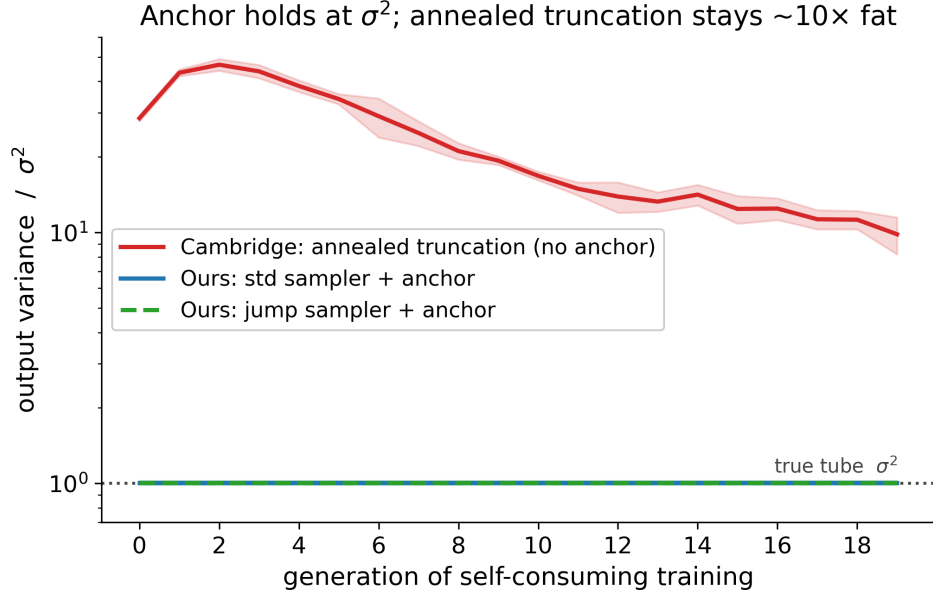


Figure 5: **Head-to-head vs. the state-of-the-art remedy (*we observe*)**. Ring manifold, $\sigma = 0.05$, $\lambda = \frac{1}{2}$, 20 generations, 5 seeds (mean $\pm$ s.d.). Red: annealed truncation $t_0: 0.05 \rightarrow 10^{-4}$, no anchor [1]—rises to $47\sigma^2$, then thins only to $9.8\sigma^2 \pm 1.7$. Blue/green: the same recursion with the anchor (standard and jump samplers): $1.01\sigma^2 \pm 0.02$, flat for 20 generations.

Dimension scaling (*we observe*). Lemma 5’s d -independence is a postdiction: across ambient $d = 8 \rightarrow 128$ (curved 2-manifolds, fixed tube), the *raw* one-shot floors grow near-linearly ($\sim d^{0.76}$ standard, $\sim d^{0.95}$ jump—the collapse itself), while the *matched* floors are flat ($\sim d^{0.1}$), and anchored recursions are flat for 8 generations at both $d = 8$ and $d = 128$.

7 Experiments

All experiments are CPU-only (≤ 12 threads), use small MLPs (3 hidden layers, width 256, SiLU, sinusoidal time embedding), denoising score matching, and geometric reverse-time grids; every long run streams to an append-only log and is exactly resumable. Full protocols, seeds, and a script-to-claim map are in the released code (Section 9).

Protocols. *Ring*: radius 2.5, $\sigma = 0.05$ ($\sigma^2 = 0.0025$), 4,000 samples/generation, 2,000 training steps. *Head-to-head* (Figure 5): $G = 20$, $\lambda = \frac{1}{2}$, 5 seeds; arm A anneals $t_0: 0.05 \rightarrow 10^{-4}$ geometrically (the remedy of 1); arms B/C add the anchor to the standard/jump samplers. *Two-horned ablation* (Figure 2): fixed $t_0 = 10^{-4}$, noise on/zeroed, $G = 15$, 3 seeds. *Pixel MNIST* (Figure 6): 8×8 pixels, $G = 8$, 3 seeds, local matcher with $k = 96$, $k_{\text{dim}} = 12$. *Law identification*: fixed- t_0 recursions at $t_0 \in \{0.02, 0.05\}$, $\lambda \in \{0.25, 0.5, 0.75\}$; the affine law fitted at one t_0 predicts fixed points at the other within 1–13% (Assumption 1 validation), and the measured excess-response slope runs $+0.6 \rightarrow \approx 0$ across pool widths (Assumption 2 validation), the net of a measured deficit channel -0.68 and injection channel $+0.73$.

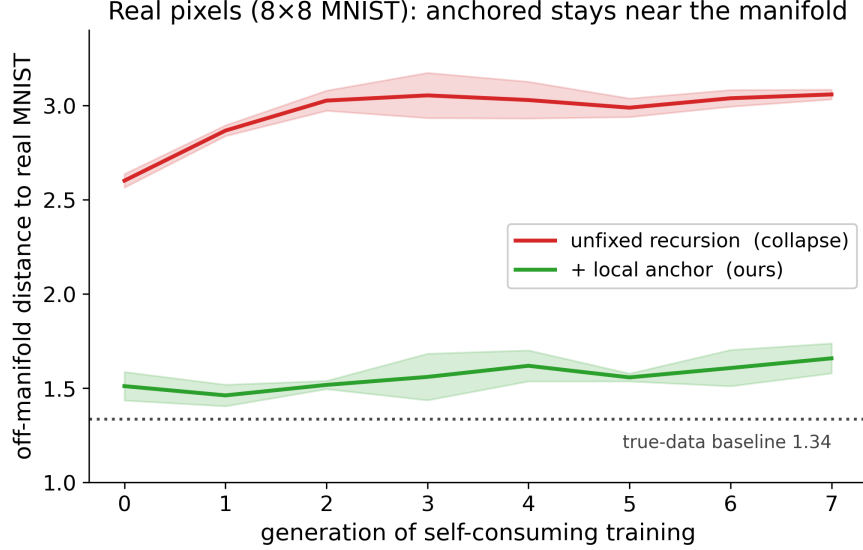


Figure 6: **Real pixels (*we observe*)**. Self-consuming recursion on 8×8 MNIST (64-dimensional pixel space), $\lambda = \frac{1}{2}$, 8 generations, 3 seeds. Off-manifold distance = mean nearest-neighbor distance to a held-out real test set. Unanchored drifts to 3.06 ($2.3\times$ the true-data baseline 1.34); anchored holds 1.66 ($1.24\times$), closing $\sim 81\%$ of the gap. The mild upward creep of the anchored arm reflects the local-chart approximation on a genuinely curved manifold.

8 Limitations and scope

Everything in the *we model* tier is collected here. (1) Assumptions 1–2 are measured, cross-validated laws about the trained pipeline, not derived facts; we present them as such, in the spirit of measured critical exponents. (2) The recursion floor constant is open: our closed forms are exact for a single pass, and lower bounds for the recursion; the measured recursion floor sits a factor 2.4 above the iterated single-pass bound (per-generation compounding, not modeled). (3) The residual-field ($\sim 30\%$ single-pass) channel is bounded in a block-correlation model (Appendix E); negative association across blocks is the one direction that could weaken that particular bound (the closed-form floor of Theorem 2 does not depend on it). (4) Curvature is carried at order $O(s^2/\tau)$ in the flat-tube idealization; sphere experiments bound the resulting bias empirically (non-compounding), but we do not compute the constants. (5) The anchor controls normal *second moments*; tails and tangential statistics improve but are not anchored—on strongly curved data the anchored recursion shows a mild residual creep (Figure 6). (6) A parameter-free one-step prediction of MNIST-latent thin ratios fails quantitatively (it predicts the direction and the regime boundary, not the number); we flag it rather than fit it. (7) Our evidence is deliberately small-scale and fully instrumented: a ring, curved 2-manifolds swept across ambient dimension $d = 8 \rightarrow 128$, six (t_0, λ) law-identification cells, the two-horned ablation, and pixel-space MNIST—chosen so that every quantity the theory names is directly measurable. We do not test CIFAR-scale or latent-diffusion pipelines. The theory predicts the mechanism transfers, because the floor depends only on the *local* normal slope and not on the ambient dimension (Lemma 5, and the measured d -independence of the matched floors); confirming this at scale is the clearest next step and is left as future work.

Table 1: Headline numbers (mean $\pm$ s.d. over seeds; variance in units of σ^2 except MNIST, which reports off-manifold distance).

Experiment	Arm	Final value	Seeds
Head-to-head, $G=20$	annealed truncation [1]	9.84 ± 1.7	5
	standard sampler + anchor	1.01 ± 0.02	5
	jump sampler + anchor	1.01 ± 0.02	5
Ablation, fixed $t_0=10^{-4}$	stochastic (noise on)	10.8–13.2	3
	deterministic (noise zeroed)	0 (point collapse)	3
Pixel MNIST, $G=8$	unanchored	3.06	3
	anchored	1.66	3
	true-data baseline	1.34	—
Floor prediction	measured $\bar{\kappa}=3.9 \Rightarrow \Phi_{\text{det}}$	3.8 (zero fit)	—

9 Reproducibility

Every claim maps to a script, an append-only result log, and a figure in the released code (ring, ablation, MNIST, floor law, law identification; ~ 2 CPU-hours total). The four result figures regenerate from logged data in seconds. A README lists the run order, the twelve load-bearing scripts, and the headline numbers a reproduction must match.

Acknowledgments

Numerical experiments, figure preparation, and portions of the derivations were carried out with the assistance of an AI system (Claude, Anthropic), used as a research tool under the author’s direction; the author reviewed all mathematics and takes full responsibility for every claim.

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A Proof of Theorem 1

Write $g(v) := T(v) - v$ with T as in (2); g is concave (composition of an affine map with the concave nondecreasing $w \mapsto w + e^2(w)$, minus a linear term), with $g(0) = T(0) = \lambda\sigma^2 + e^2(\lambda\sigma^2) > 0$ and one-sided derivative $g'(v) = (1 - \lambda)(1 + e^{2'}(w(v))) - 1$, nonincreasing in v with limit $(1 - \lambda)(1 + c_\infty) - 1 < 0$ when $\lambda > \lambda^*$.

(i) Existence, uniqueness. A concave function positive at 0 whose derivative is eventually $\leq -\delta < 0$ tends to $-\infty$, so g has a zero; concavity plus $g(0) > 0$ forbids two downcrossings, so the zero v_∞ is unique, with $T(v) > v$ for $v < v_\infty$ and $T(v) < v$ for $v > v_\infty$.

Contraction above. For $v \geq v_\infty$, monotonicity and concavity give $0 \leq T(v) - v_\infty \leq \rho_{\text{loc}}(v - v_\infty)$ with $\rho_{\text{loc}} = T'(v_\infty^+) < 1$.

Convergence under vanishing perturbations. Let $v_g = T(v_{g-1}) + s_g^2$, $s_g^2 \rightarrow 0$. While iterates stay $\geq v_\infty$, $x_g := v_g - v_\infty \leq \rho_{\text{loc}}x_{g-1} + s_g^2 \rightarrow 0$ by the standard perturbed contraction argument (split $\sum_j \rho^{g-j} s_j^2$ at $j = g/2$). If an iterate drops below v_∞ : define the unperturbed comparison $u_g = T(u_{g-1})$ started there; then $u_g \uparrow v_\infty$ (monotone, bounded, $T(v) > v$ below v_∞) and $v_g \geq u_g$ by induction ($s_g^2 \geq 0$, T monotone), so $\liminf v_g \geq v_\infty$, while $v_g < v_\infty + s_g^2$ gives $\limsup v_g \leq v_\infty$. Alternating regimes inherit both barriers. Linearizing T at v_∞ gives the local rate.

(ii) Divergence. If $(1 - \lambda)(1 + c_\infty) > 1$ then, since $e^{2'} \geq c_\infty$ everywhere (the concave derivative decreases to its limit), $g' \geq (1 - \lambda)(1 + c_\infty) - 1 > 0$, so g is nondecreasing and $g(v) \geq g(0) > 0$ for all v : $v_g \geq v_0 + g \cdot T(0) \rightarrow \infty$, regardless of the schedule ($s_g^2 \geq 0$ only helps).

(iii) Substituting $e^2(w) = e_0^2 + cw$ makes T affine with slope $(1 + c)(1 - \lambda)$; solve the fixed point. $\square$

B The variance ODE and the orthogonality shield

Lemma 1. One Euler step $x' = (1 + (\frac{1}{2} - \hat{\kappa}/s) dt)x - (b/s) dt + \sqrt{dt} \xi$ is affine in x plus independent noise, so $\text{Var}(x') = (1 + (\frac{1}{2} - \hat{\kappa}/s) dt)^2 V + dt = V + [(1 - 2\hat{\kappa}/s)V + 1] dt + O(dt^2)$; let $dt \rightarrow 0$. With $\hat{\kappa} = \kappa = s/\gamma^2$: $\kappa/s = 1/\gamma^2$ and $V = \gamma^2$ solves the equation exactly (direct check), confirming the bookkeeping. The intercept b moves only the mean. $\square$

Lemma 2. Define $(\hat{\kappa}, b)$ as the affine $L^2(p_t)$ projection of the estimator itself; the residual $\delta = \hat{\varepsilon} - \hat{\kappa}x - b$ satisfies the normal equations $\mathbb{E}_{p_t}[\delta] = \mathbb{E}_{p_t}[x\delta] = 0$ by definition. The first variation of the variance flow under drift perturbation $\eta \delta/s$ is $(2/s)\mathbb{E}_{p_t}[x\delta] dt = 0$ at $\eta = 0$: no first-order thinning. The leading effect is the accumulated squared displacement $\mathbb{E}[u^2] \geq 0$ of the integrated kicks δ/s —injection. (Along the reverse path the ensemble deviates from p_t ; the cross term is then $O(\|\hat{p}_t - p_t\| r(t))$, second order in the total perturbation.) $\square$

C Floor lemmas and closed forms

Lemma 3. From (1), $\kappa(t) = s/(a^2\sigma^2 + s^2)$ with $s^2 = 1 - e^{-t}$: $\kappa \rightarrow 0$ as $t \rightarrow 0$ ($s \rightarrow 0$, $\gamma^2 \rightarrow \sigma^2$), $\kappa \approx 1/s$ for $s^2 \gg \sigma^2$, and maximizing over s gives $\kappa_{\text{max}} = 1/(2\sigma)$ at $s = \sigma$ (i.e. $t \approx \sigma^2$), to leading order in σ^2 . Below the band the remaining e-folds are $\int_{t_0}^{t_-} 2\kappa/s dt = \int 2 dt/\gamma^2 = 2 \log \frac{\sigma^2 + t_-}{\sigma^2 + t_0} \leq 2 \log(1 + \rho^2/4) \leq \rho^2/2$, using $t_- \approx \bar{\kappa}^2 \sigma^4$. $\square$

Lemma 4. Substitute $t = \sigma^2 u$, $V = \sigma^2 W$: $s^2 = \sigma^2 u(1 + O(\sigma^2 u))$, $\gamma^2 = \sigma^2(1 + u)(1 + O(\sigma^2 u))$, so $\kappa = (1/\sigma) \frac{\sqrt{u}}{1+u}(1 + O(\sigma^2 u))$ and $2\hat{\kappa}/s = (2m(u)/\sqrt{u})/\sigma^2$; the ODE becomes $-dW/du = 1 - (2m/\sqrt{u})W + \sigma^2 W(1 + o(1))$. On $u \in [0, U]$ the perturbing terms are $O(\sigma^2 U)$; the band

occupies $u \leq u_+ = O(1/\rho^2)$, giving the stated $O(\sigma^2/\rho^2)$. For the boundary: the uncapped equation $-dW/du = 1 - 2W/(1+u)$ has general solution $W = (1+u) + A(1+u)^2$ (direct check), so deviations from the slave solution $W = 1+u$ contract as $((1+u)/(1+u'))^2$ under downward integration from u' to u ; the influence of the $t \gtrsim 1$ boundary layer at band entry is suppressed by this quadratic factor and is absorbed in the same error term. *Numerical check:* the slave solution is invariant to 3×10^{-15} under the integrator; finite- σ floors converge to the limit values from above (e.g. $\rho = 0.39$: 3.953/3.812/3.775 at $\sigma = 0.10/0.05/0.02$ vs. limit 3.769). $\square$

Theorem 2(U). By the scalar comparison principle (equal forcing and data; smaller $\hat{\kappa}$ means larger V), any profile with $\hat{\kappa} \leq \bar{\kappa}$ has $V \geq V_{\hat{\kappa} \equiv \bar{\kappa}}$, one-sided—no lower bound on $\hat{\kappa}$ is used. For the constant envelope, (4) with $m \equiv \rho/2$ has integrating factor $e^{-2\rho\sqrt{u}}$: $\frac{d}{du}[We^{-2\rho\sqrt{u}}] = -e^{-2\rho\sqrt{u}}$, so for data of sub-quadratic growth at $u = \infty$, $W(0^+) = \int_0^\infty e^{-2\rho\sqrt{u}} du = 2 \int_0^\infty xe^{-2\rho x} dx = \frac{1}{2\rho^2}$, exactly, for every $\rho > 0$. $1/(2\rho^2) > 1 \iff \rho < 1/\sqrt{2}$.

Theorem 2(N). The band edges solve $\sqrt{u}/(1+u) = \rho/2$, a quadratic in $x = \sqrt{u}$: $\rho x^2 - 2x + \rho = 0$, $x_\mp = (1 \mp q)/\rho$, $q = \sqrt{1 - \rho^2}$ (nonempty iff $\rho < 1$). Above the band the solution is the exact slave $W = 1+u$; it enters the band with $W(u_+) = 1+u_+$. Across the band ($m = \rho/2$), the same integrating factor with antiderivative $F(u) = -(2\rho\sqrt{u} + 1)e^{-2\rho\sqrt{u}}/(2\rho^2)$ gives

$$W(u_-) = (1 + u_+)e^{-2\rho(x_+ - x_-)} + \frac{(2\rho x_- + 1) - (2\rho x_+ + 1)e^{-2\rho(x_+ - x_-)}}{2\rho^2},$$

and $2\rho x_\mp = 2(1 \mp q)$, $2\rho(x_+ - x_-) = 4q$ yield the displayed W_{exit} . Below the band the equation is uncapped with exact general solution $(1+u) + A(1+u)^2$; matching at u_- gives $A = (W(u_-) - (1+u_-))/(1+u_-)^2$ and $g(\rho) = W(0^+) = 1 + A$, which is (6). Limits: as $\rho \rightarrow 1^-$ ($q \rightarrow 0$), $u_\pm \rightarrow 1$ and $g \rightarrow 1$; as $\rho \rightarrow 0$ ($q \rightarrow 1$), $u_+ \approx 4/\rho^2$, $u_- \approx \rho^2/4$, and expanding, $g = [8e^{-4} + 1 - 5e^{-4}]/(2\rho^2) + O(1) = (1 + 3e^{-4})/(2\rho^2) + O(1)$, giving $C_0 = (1 + 3e^{-4})/2$. Positivity $g > 1$ on $(0, 1)$ follows from $W(u_-) > 1 + u_-$ (the band under-contracts relative to the slave). *Numerical check:* (6) and the exact $1/(2\rho^2)$ match direct integration of (4) to $\leq 2.4 \times 10^{-5}$ over $\rho \in [0.1, 1]$ (pure Euler residual).

Finally, Lemma 2 extends both bounds from the affine part to the full estimator (the residual only adds variance), and Lemma 4 transfers the limit values to finite σ with the stated error. $\square$

Corollary 1 is Theorem 2(U) read at fixed $\bar{\kappa}$ with $\sigma \rightarrow 0$; the capacity bound inverts $1/(2\rho^2) \leq C$. $\square$

D Posterior-mean sharpening and per-axis necessity

Proposition 1 (Tweedie sharpening; we prove). *In the Gaussian normal coordinate with axis variance σ_{ax}^2 , the jump with the exact predictor outputs the posterior mean, with*

$$\text{Var}(\hat{x}_0) = \frac{a^2 \sigma_{\text{ax}}^4}{\gamma_{\text{ax}}^2(\hat{t})}, \quad \frac{\text{Var}(\hat{x}_0)}{\sigma_{\text{ax}}^2} = \frac{a^2 \sigma_{\text{ax}}^2}{a^2 \sigma_{\text{ax}}^2 + s^2(\hat{t})} < 1,$$

and with a trained predictor $\hat{\varepsilon} = \varepsilon^* - \tilde{\delta}$, $v_{\text{jump,ax}} = a^2 \sigma_{\text{ax}}^4 / \gamma_{\text{ax}}^2 + (s^2/a^2) \mathbb{E}[\tilde{\delta}^2] + 2(s/a) \text{Cov} + O(s^4/\tau^2)$.

Proof. $\hat{x}_0 = (x - s\varepsilon^*)/a = x(1 - s^2/\gamma^2)/a = (a\sigma_{\text{ax}}^2/\gamma^2)x$, so $\text{Var} = a^2 \sigma_{\text{ax}}^4/\gamma^2$; Tweedie's formula identifies this with $\mathbb{E}[X_0 | X_{\hat{t}}]$ [14]. The error term is the jump map applied to $\varepsilon^* - \tilde{\delta}$; the curvature term is the posterior-mean chord bias. $\square$

Consequences: strongly singular axes ($\sigma_{\text{ax}} \ll s$) are collapsed onto the manifold (the matcher must top up); fat axes ($\sigma_{\text{ax}} \gg s$) are untouched while injection fattens them (the matcher must contract); axes with $\sigma_{\text{ax}} \sim s$ sharpen by an axis-dependent factor strictly inside $(0, 1)$. Hence

the required correction is signed and anisotropic: any correct matcher must be bidirectional and per-axis. (*We observe:* scalar matching leaves thin ratios at 0.62–0.65 on MNIST latents; per-axis matching achieves $|\text{thin} - 1| \leq 0.012$.)

E The residual channel (*we model*)

The orthogonal residual δ has measured band risk $r_{\text{res}}^2(t) \approx 0.02\text{--}0.05$, flat in t and in sample size, with path-coherence time $\approx \nu t$, $\nu \approx 0.75$. Under a block-correlation model (coherent within blocks of length νt , independent across blocks, no systematic over-contraction), the injected variance obeys $\Phi_0(t_0) \geq c \nu \rho_0(\sigma^2 + t_0)$: decreasing in t_0 until $t_0 \sim \sigma^2$, then saturating—the same shape as the measured excess (knee at $t_0 \approx \sigma^2$, n -flat), with the single constant effective to within a factor ~ 2 . In this paper the channel plays a supporting role only: it is the measured $\sim 30\%$ single-pass remainder on top of the closed-form floor, which does not depend on it.

F Experimental details

Networks and training. 3-layer MLPs, width 256, SiLU, 32-dim sinusoidal time embedding; Adam, learning rate 2×10^{-3} ; batch 512 (ring) / 256 (MNIST); denoising score matching with $t \sim U(10^{-3}, 8)$; fresh network each generation.

Sampling. Reverse Euler–Maruyama on a 200-point geometric grid from $T = 8$; the ablation zeroes the $\sqrt{|dt|} \xi$ term only. Jump: $\hat{t} = 0.05$.

Anchor. Ring: radial (normal) bidirectional match of the pool’s radial variance to the stale reference’s, contract-by-scaling / top-up-by-noise. MNIST: per-point k NN-PCA ($k = 96$, $k_{\text{dim}} = 12$) charts from the reference; normal coordinates rescaled to the reference’s local normal eigenvalues in the same frames. Reference: $m = 2,000$ real samples drawn once at $g = 0$, never refreshed (provenance-free: the matcher is applied to the whole pool without labels).

Measured network profile. The fitted slope $\hat{\kappa}(t)$ is extracted by affine regression of $\hat{\varepsilon}$ against x on fiber lines at each t ; it tracks $\kappa(t)$ (0.95–0.97 for $t \geq 0.1$), saturates at $\bar{\kappa} \approx 3.9$ inside the band, and never exceeds κ (n -independent across 1k–16k). The decomposition identity $(\kappa - \hat{\kappa})^2 \gamma^2 + r_{\text{res}}^2 = r_{\text{tot}}^2$ checks numerically along the path.

Two-channel anatomy of c . Training single-width tubes at $w \in \{0.0009, \dots, 0.0169\}$ and measuring $dv/dw - 1$ gives net $c \in \{0.00, 0.05, 0.13, 0.10, 0.05\}$ —small, positive, concave—as the sum of a deficit response ≈ -0.68 (a fatter pool needs a smaller peak slope) and an injection response $\approx +0.73$ (a fatter pool damps injected variance less).